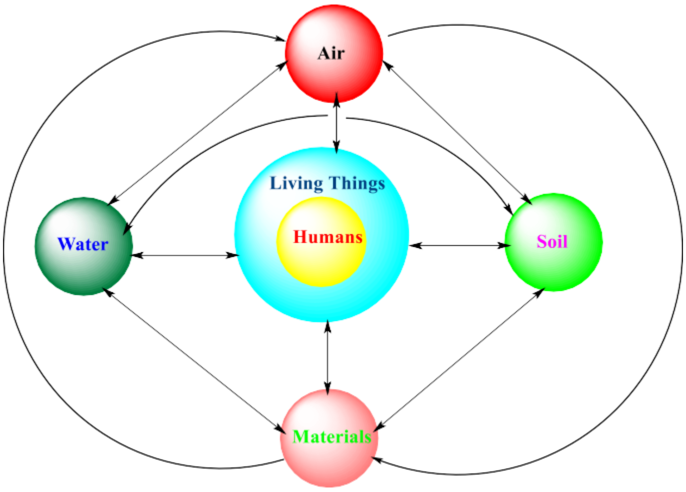
Unit-I

Esosystem

Environment is derived from the word "Environner" which mean “encircle” or Surroundings. The environment means surroundings, i.e. air (atmosphere), water (hydrosphere). Soil (lithosphere) and living entities (biosphere).



Environment can be defined as the sum of all physical, biological, chemical and Socio-economic factors constituting the surroundings of mankind. Environmental Science is a branch of philosophy that deals with a systematized knowledge of the facts of Nature. Environment is varied from place to place, continent to continent depending upon the physiography, Topography, climate and available natural resources. To understanding the functions of various components of environment is called as “**Environmental Science**”.

Concept of an ecosystem

All living organism are exchanging matter and energy from their environment in Surroundings. How do they derive energy and nutrients to live together and how do they influence each other questions are answered by ecology.

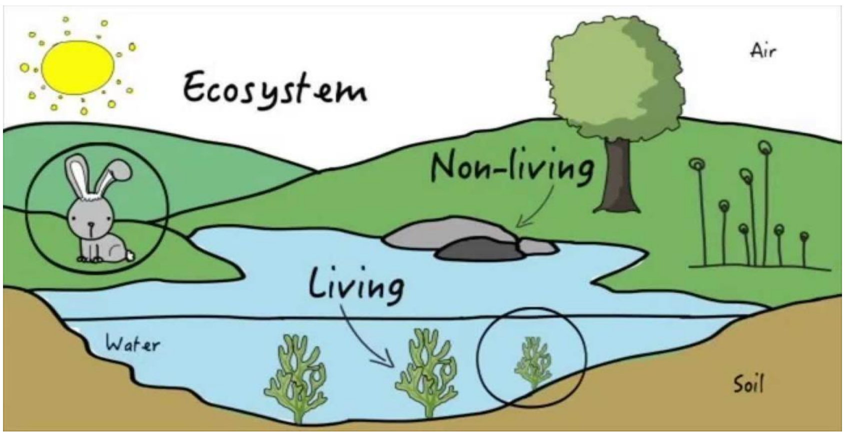


Fig.1: Ecosystem

Ecology: It deals with the study of organisms in their natural home interacting with their Surroundings.

Biotic (living) & Abiotic (non-living) surroundings Or Ecology is the study of ecosystems

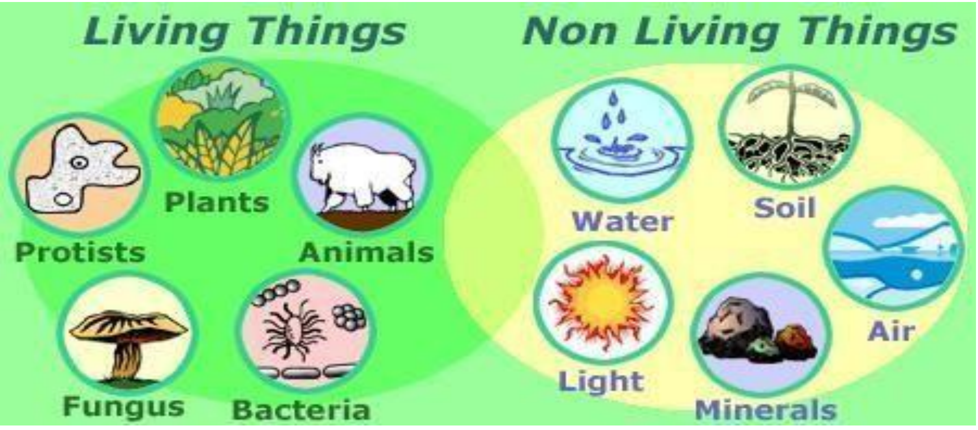


Fig.2: Living and nonliving things

Ecosystem:

Ecosystem is the basic functional unit of ecology, Ecosystem = study of home (In Greek), Ecosystem has been defined as a Systems of organisms with Surroundings

Eg: Animal cannot prepare their food; they depend upon the plants.

Scope of ecosystems:

- Helps in to tackle problems like pollution, floods, O3 depletion, global warming.
- It is necessary in maintaining ecological balance and understanding different cycles (oxygen, nitrogen, carbon etc.)
- Helps in protecting flora and fauna
- we can maintain balance in nature and can prevent ecological disaster
- played an Important role in human welfare, agriculture, conservation of wild life.
- ecology in the scientific study of interaction between organisms and the environment, which intern determine both the distribution of organisms and their abundance.

Importance of ecosystem

- Ecosystem Services are the natural services or earth capital that support life on the earth.
- Essential to the quality of human life and to this functioning of the world economics

Ecosystem Services include:

- Controlling and Moderating climate.
- Providing and renewing air, water, soil.
- Recycling vital nutrients through chemical cycling.
- providing renewable and non-renewable energy Sources and Minerals.
- Furnishing people with food, fibber, medicines timber and paper.
- Pollinating crops and other plant species.

- Absorbing, diluting and detoxifying many populations and toxic chemicals.
- Helping control of populations pests and disease organisms.
- Slowing erosion and preventing floods.
- Providing biodiversity of genes and species.

Concepts of ecosystem

In an ecosystem, the interaction of life with its environment takes place at many levels. A single bacterium in the soil interacts with water, air around it within a small space while a fish in a river interacts with water and other animals, rivals in a large space.

Considering the operational point of view; the biotic and abiotic components of an ecosystem are so interlinked such that their separation from each other is practically difficult. So, in an ecosystem both organisms (biotic communities) and abiotic environment (rainfall, temperature, humidity) each influence the properties with other for maintenance of life.

Structure or components of an ecosystem

The term refers to various components. An ecosystem has two Major component.

(I) Biotic (living) components

(II) Abiotic (non-living) components

(I) Biotic components: Living components in an ecosystem are either producers or consumers. They are also called biotic components. Producers can produce organic components

e.g. plants can produce starch, carbohydrates, cellulose from a process called **photosynthesis**.

$$6CO_2 + H_2O \longrightarrow \underset{}{6O_2} + \underset{\text{Glucose}}{C_6H_{12}O_6}$$

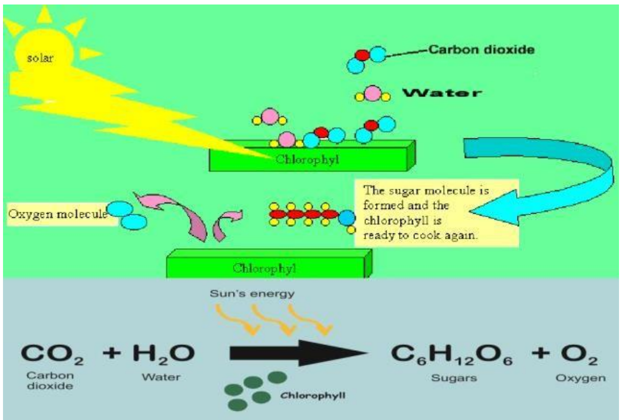


Fig.3: Photosynthesis process

Consumers are the components that are dependent on producers for their food

e.g. human beings and animals

Biotic Components are further classified into **three** main groups

1. Producers
2. Consumers

3. Decomposers or Reducers

1. **Producers (Autotroph):** The green plants have chlorophyll with the help of which they trap solar energy and change it into chemical energy of carbohydrates using simple inorganic compound namely, water and carbon dioxide. This process is known as **photosynthesis**. The chemical energy stored by the producers is utilized partly by the producers for their own growth and survival and the remaining is stored in the plants for their future use. They are classified into two categories based on their source of food.

a) **Photoautotrophs:** An organism capable of synthesizing its own food from inorganic substances using light as an energy source. Green plants and photosynthetic bacteria are photoautotrophs.

b) **Chemotrophs**: Organisms that obtain energy by the oxidation of electron donors in their environments. These molecules can be **organic** (chemoorganotrophs) or **inorganic** (chemolithotrophs).

2. **Consumers (Heterotrophs)**: The animals lack chlorophyll and are unable to synthesis their own food therefore they depend on the producers for their food.

   They are known as heterotrophs (i.e. heteros = others, trophs = feeder).

   The Consumers are of **three** types:

   (a) **Primary Consumer** (**Herbivores**): **i.e**. Animal feeding on plants,

   e.g. Rabbit, deer, goat etc.

   (b) **Secondary Consumers (Primary carnivores)**: The animal feeding on Herbivores are called as secondary Consumer or primary carnivores.

   e.g. Cats, foxes, snakes.

   (c) **Tertiary Consumers (secondary carnivores):** These are directly depending on the Secondary and primary consumers for their food.

   e.g. Tiger, Lion etc.

3. **Decomposers or Reducers**: Bacteria & fungi belong to this category. They break down the dead organic matter of producers & consumers for their food and release to the environment the simple inorganic and organic substance. These simple substances are reused by the producers resulting in a cyclic exchange of material between biotic & abiotic environment.

Eg: Bacteria, Earth worms, Beetles etc.

(II) Abiotic components: The non-living components (physical & chemical) of ecosystem collectively.

The abiotic component of the ecosystem is divided into three types

- Climatic factors: Solar energy, temperature, air etc.
- Physical factors: light fire pressure tell

- Chemical factor: Acidity, Salinity, inorganic nutrients Like C, H, O, P, K, N and organic substance like proteins, lipids, carbohydrates etc.

Functions of an Ecosystems

Eco-systems have some functional attributes due to which components remain and running together. The tendency of every eco-system depends on various function performed by the structural components of the eco-system.

Functions of an ecosystem are of **three** types.

1. **Primary Function**: The primary function of ecosystems is manufacturing of starch by photosynthesis.
2. **Secondary function**: The secondary function of an ecosystem is distributing the energy in the form of food for all consumers.
3. **Tertiary function:** All living systems are died at a particular stage. These dead systems are decomposed that is the third function of ecosystem.

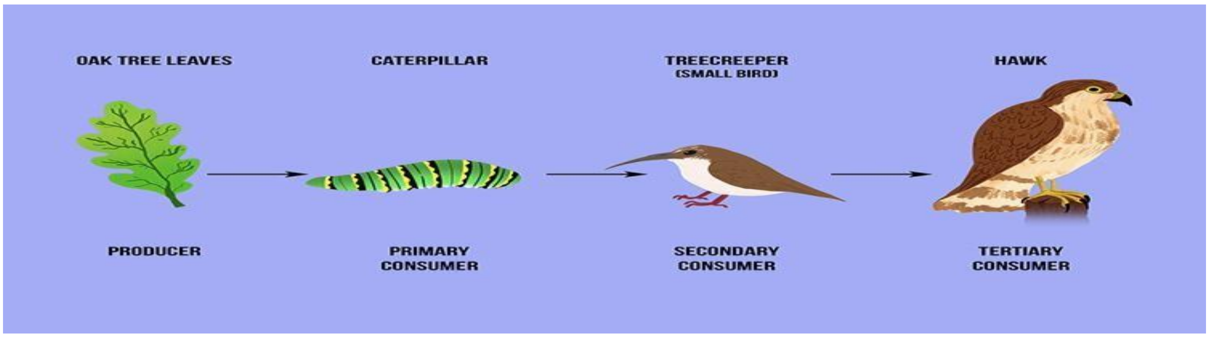


Fig.4: Function of an ecosystem

Energy flow in the ecosystems

Biological activities require energy which ultimately comes from the sun. Solar energy is transformed into chemical energy by a process of photosynthesis this energy is stored in plant tissue and then transformed into heat energy during metabolic activities. Thus, in biological world the energy flows from the sun to plants and then to all heterotrophic organisms. The flow of energy is unidirectional and non-cyclic.

This one-way flow of energy is governed by laws of thermodynamics which states that:

(a) Energy can neither be created nor be destroyed but may be transformed from one form to another.

(b) During the energy transfer there is degradation of energy from a concentrated form (mechanical, chemical, or electrical etc.) to a dispersed form (heat).

No energy transformation is 100 % efficient; it is always accompanied by some dispersion or loss of energy in the form heat. Therefore, biological systems including ecosystems must be supplied with energy on a continuous Basis.

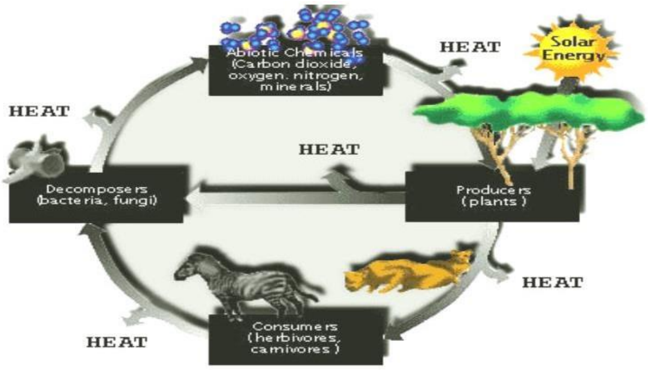


Fig.5: Energy flow in the ecosystem

The flow of energy in an ecosystem follows the laws of thermodynamics.

I law of thermodynamics - "Energy neither can be created nor destroyed, but it can be converted from one form to other".

Energy for an ecosystem comes from the sun. It is absorbed by plants; it is converted into chemical energy. This chemical energy utilized by consumers transform into heat.

II law of thermodynamics - "Whenever energy is transformed, there is a loss of energy through the release of heat".

Energy is transferred between tropic levels in the form of heat as it moves from one tropic level to another tropic level.

The loss of energy takes place through work, running, hunting etc.,

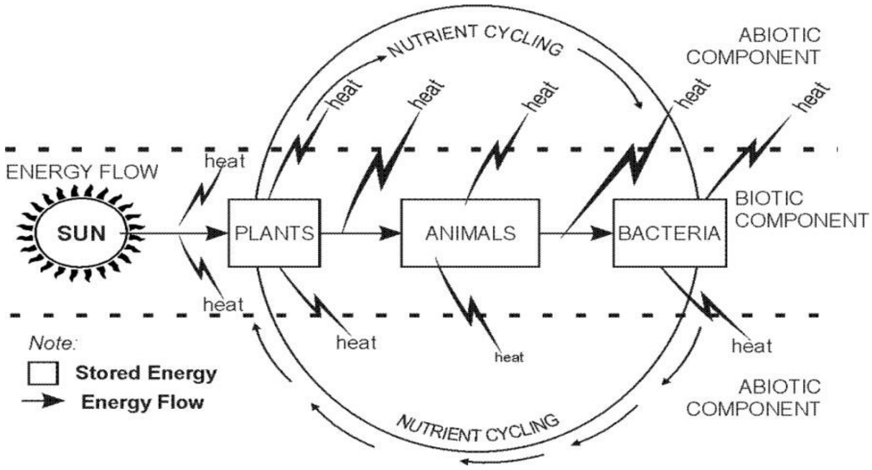


Fig.6: Flow of energy and nutrient cycling from abiotic to biotic and vice versa.

<u>Food chain:</u>

There sequence of eating and being eaten in an ecosystem is known as "food chain"

(or)

"Transfer of food energy from the plants through a series of organisms is known as food chain"

A food chain always starts with plant life and ends with animal. When the organisms die, they are all decomposed by microorganism (bacteria and fungi) into nutrients that can again be used by the plants. At each and every level, nearly 80-90% of the potential energy gets lost as heat.

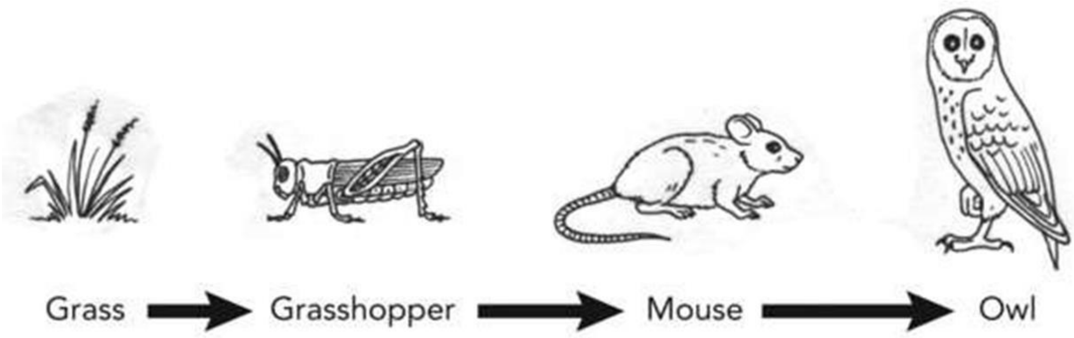


Fig.7: General food chain process.

Tropic Levels or Feeding levels

The various steps through which food energy passes in an ecosystem is called as tropic levels. The tropic levels are arranged in the following way,

- The green plants or producers represent first tropic level T1,
- The primary consumers represent second tropic level T2.
- The secondary consumers represent third tropic level T3.
- The tertiary consumers are fourth tropic level T4.
- Finally, decomposers represent last tropic level T5.

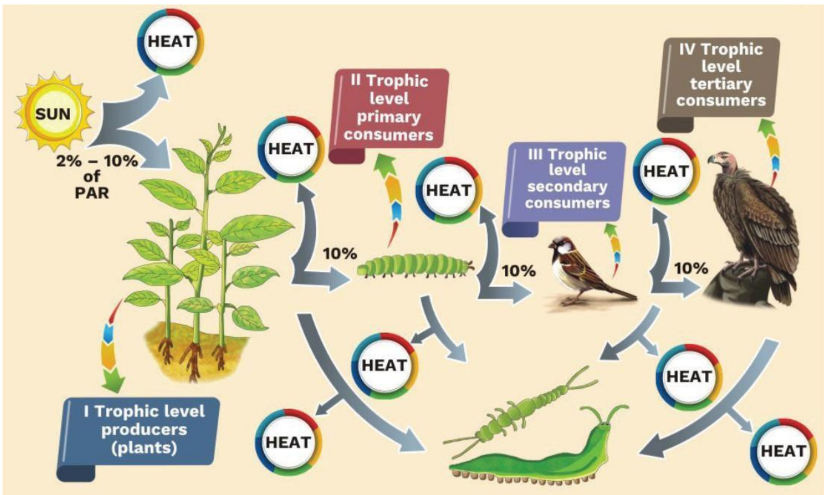


Fig.8: Energy flow through different trophic levels.

Types of Food Chains: There are two types of food chains

1. Grazing food chain
2. Detritus food chain

1. **Grazing food chain**: Found in Grassland ecosystems and pond ecosystems. Grazing food chain starts with green plants (primary producers) and goes to decomposer food chain or detritus food chain through herbivores and carnivores.

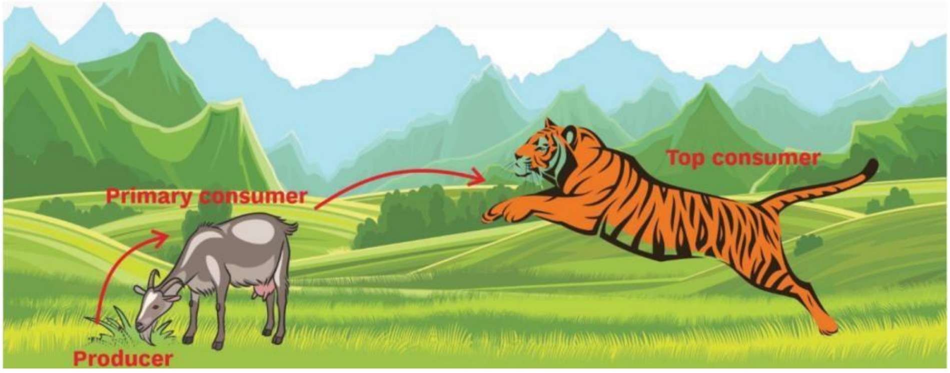


Fig.9: Grazing food chain

2. **Detritus food chain**: Found in Grassland ecosystems and forest ecosystems. Detritus food chain starts with dead organic matter (plants and animals) and goes to decomposer food chain through herbivores and carnivores.

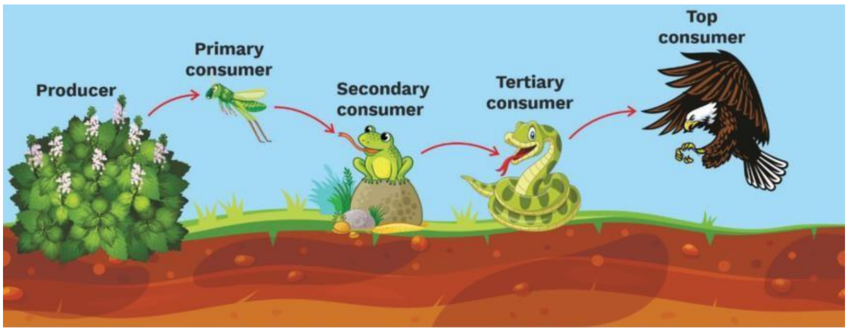


Fig: 10: Terrestrial food chain

<u>Food web</u>

The interlocking pattern of various food chains in an ecosystem is known as food web. In a food web many food chains are interconnected, where different types of organisms are connected at different tropic levels, so that there are a number of opportunities of eating and being eaten at each tropic level.

Grass may be eaten by rabbit, rats, deer's, etc., these may be eaten by carnivores (snake, fox, tiger).

Thus, there is an interlocking of various food chains called food webs.

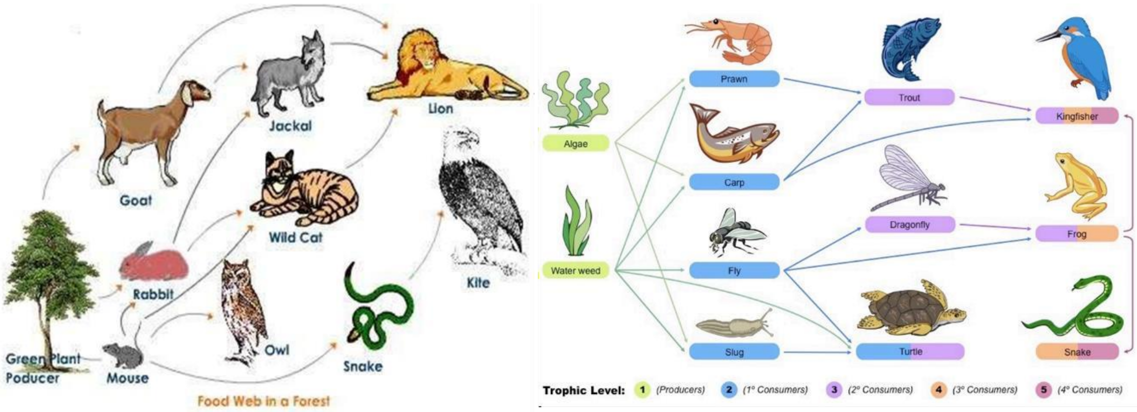


Fig: 11: food web in a forest and energy flow through different trophic levels

Significance of food chains and food webs

- Food chains and food webs play a very important role in the ecosystem. Energy flow and nutrient cycling takes place through them.
- They maintain and regulate the population size of different tropic levels, and thus help in maintaining ecological balance.
- They have the property of bio-magnification. The non – biodegradable materials keep on passing from one tropic level to another.
- At each successive tropic level, the concentration keeps on increasing.
- This process is known as bio-magnification.

Ecological pyramids:

The different species in a food chain are called tropic levels. Each food chain has 3 mains trophic level, producer, consumer, and decomposers. Thus, Graphical representation of these trophic levels is called as **Ecological Pyramids**. It was devised by an ecologist "**Charles Elton**" therefore this pyramid is also called **Ecological pyramid** or **Eltonian pyramids**.

Types of Ecological Pyramids:

Ecological pyramids are of three types: **1. Pyramid of numbers.**
2. Pyramid of energy.
3. Pyramid of biomass.

1. Pyramid of numbers:

It represents the number of individual organisms present in each tropic level.

E.g. A grassland Ecosystem

Producers are grass (small in size and large in number. Hence, they occupy the first tropic level. The primary consumers are rats occupying the second tropic level. It is worthwhile to note that rats are less in number than grass. Secondary consumers are snakes which occupy the third tropic level and they are lesser in number than rats. Tertiary consumers are Eagles that occupy the next tropic level. This is the last tropic level where the number and size of the tropic level is the least as shown in the diagram.

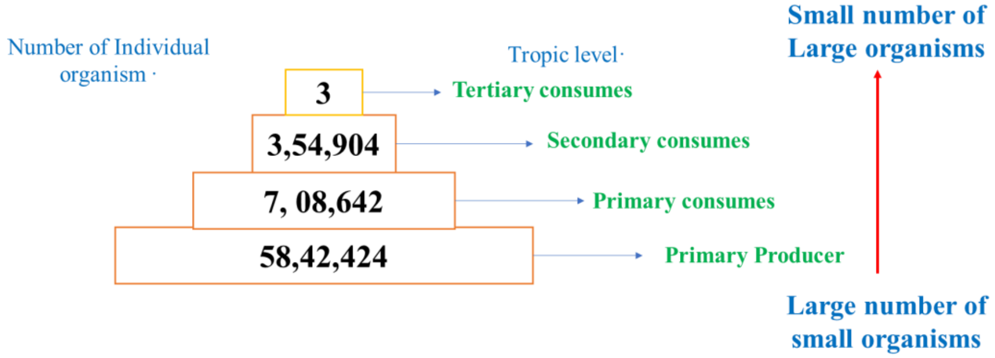


Fig.12: Pyramid of No. in a grassland ecosystem. Only three top-carnivores are supported in an ecosystem based on production of nearly 6 million plants.

2. Pyramid of energy:

It represents the amount of energy present in each tropic level. The rate of energy flow and the productivity at each successive tropic level is shown in the figure below. At every successive tropic level, there is a heavy loss of energy (almost 90%) in the form of heat.

Thus, at each tropic level only 10% is transferred. Hence there is a sharp decrease in energy at each and every successive tropic level as we move from producers to top consumers (carnivores). Pyramid of energy is depicted in the figure below.

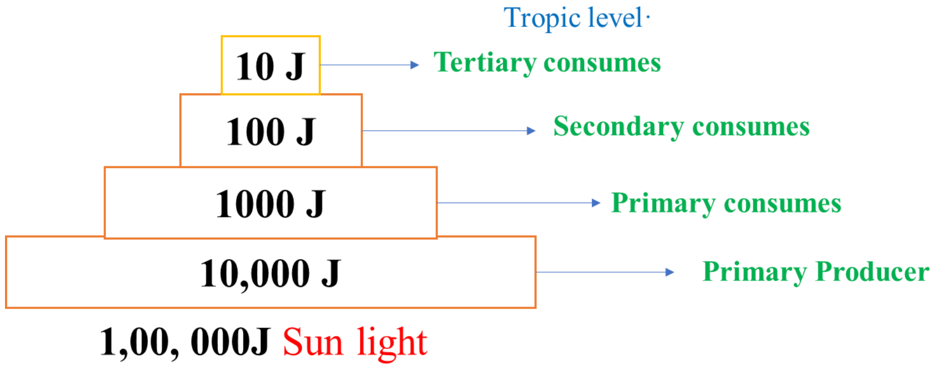


Fig.13: Pyramid of energy

3. Pyramid of biomass:

It represents the total amount of biomass (mass or weight of biological material) present in each tropic level. Considering the example of a forest ecosystem, there is a steady decrease in the biomass from the lower tropic level to the higher tropic level. The producers (trees) contribute a major amount of the biomass. The next tropic levels are the herbivores (insects and birds) and carnivores (snakes, foxes, etc). The top of the tropic level consists of very few tertiary consumers (Ex: Lions and Tigers) whose biomass is very low. The pyramid of biomass is shown below.

Fig.13: Pyramid of biomass shows a sharp decrease in biomass at higher trophic levels.

Classification of ecosystems

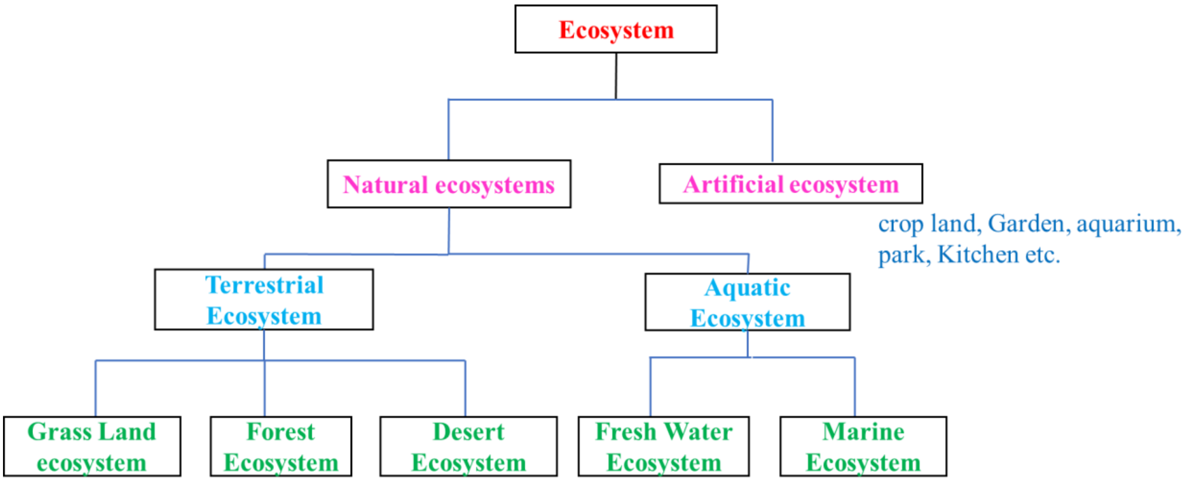
Due to the abiotic factors, different ecosystems develop in different ways. These factors and their interaction between each other and with biotic components have resulted in formation of different types of ecosystems as explained below. Ecosystem may be natural or artificial.

Artificial Ecosystem: These are maintained or created artificially by man. The man tries to control biotic community as well as physico-chemical environment.

Eg: Artificial Pond, urban area development.

Natural Ecosystem:

It consists of Terrestrial and Aquatic Ecosystems which are maintained naturally.



Terrestrial Ecosystem

Terrestrial ecosystems are many because there are so many different sorts of places on Earth.

Some of the most common terrestrial ecosystems that are found are the following:

1. Grassland ecosystem
2. Forest ecosystem
3. Desert ecosystem

1. Grass land ecosystem:

These occupy a comparatively fewer area, roughly 20% of the earth's surface.

The various components of grassland ecosystem are:

Abiotic components: These include the nutrients present in soil and the aerial environment. C, H, O, N, P, S are supplied by Carbon dioxide, water, nitrates, phosphates and sulphates.

Biotic components: These may be categorized as:

Producers: These are mainly grasses e.g. Cynodont species, Dicanthium species etc. Shrubs may also be present.

Consumers: These occur in the following sequence:

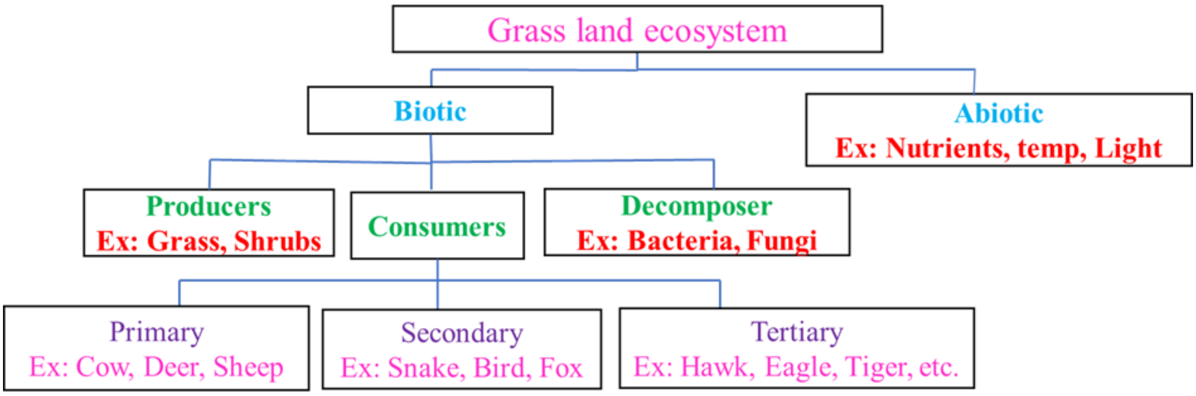
Primary consumers are the herbivores feeding on grasses are mainly grazing animals as cows, deer's and rabbit etc. Besides them there are insects, termites and millipedes that feed on leaves.

Secondary consumers: Carnivores feeding on herbivores **e.g**. Fox, Snakes, frogs, Lizards etc.

Decomposers: Microbes including fungi like Mucor, Aspergillus, Rhizopus etc and some bacteria and actinomycetes.

Characteristics:

- Grass land ecosystem is a plain land occupied by grasses.
- Soil is very rich in nutrients and organic matter
- It is characterized by low as uneven rainfall.
- Since it has tall grass, it is ideal place for grazing animal.



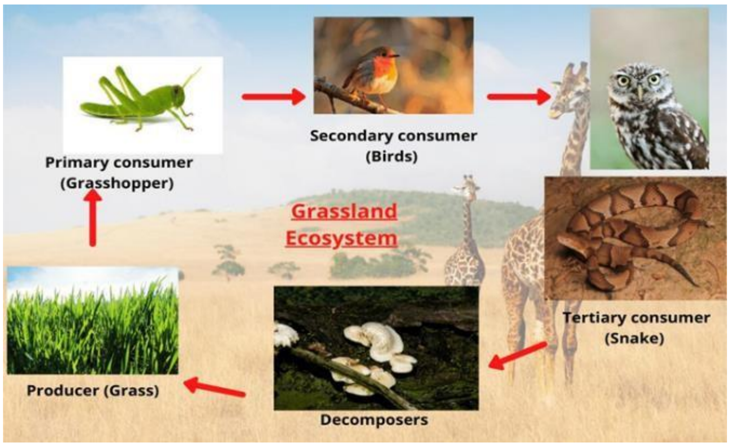


Fig.14: Structure & functions of Grass land of ecosystem

2. Forest ecosystem:

Forests occupy roughly 40 % of the land. In India these occupy roughly one-tenth of the total land area. The different components of forest ecosystem include:

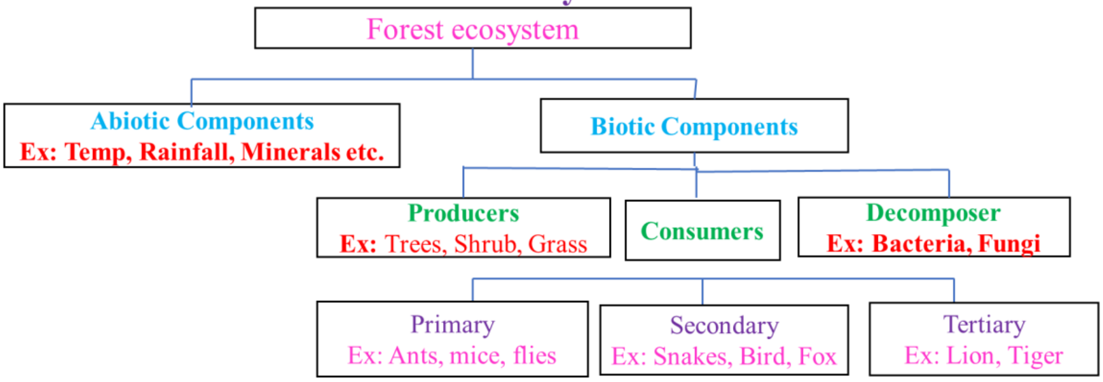
Abiotic components: Include inorganic and organic substances present in soil and atmosphere. Dead organic debris is also present in forests.

Biotic components: These include producers like trees, shrubs and ground vegetation. Among the primary consumers are the herbivores which include animals feeding on tree leaves as ants, flies, beetles, spiders etc. and larger animals like elephants, deer etc. Secondary consumers are the carnivores like snakes, birds and fox etc. Tertiary consumers are the top carnivores like lion, tiger etc. Decomposers include microorganisms like fungi, bacteria and actinomycetes.

Characteristics of Forest:

- A forest is a living community of various species of trees and small form of vegetation characterized by warm temp and adequate rainfall.
- it regulates the climate and rainfall.
- rich in organic matter and nutrient of forest soils.
- There are four types of forests: Deciduous, Tropical, coastal and coniferous forests.

Structure & functions of Forest ecosystem:



3. Deserts ecosystem:

Desert occupy about 17% of land, occurring in the regions with an annual rainfall of about 25 centimeters. The species composition of such an ecosystem is varied and typical due to the extremes of temperature and water factors.

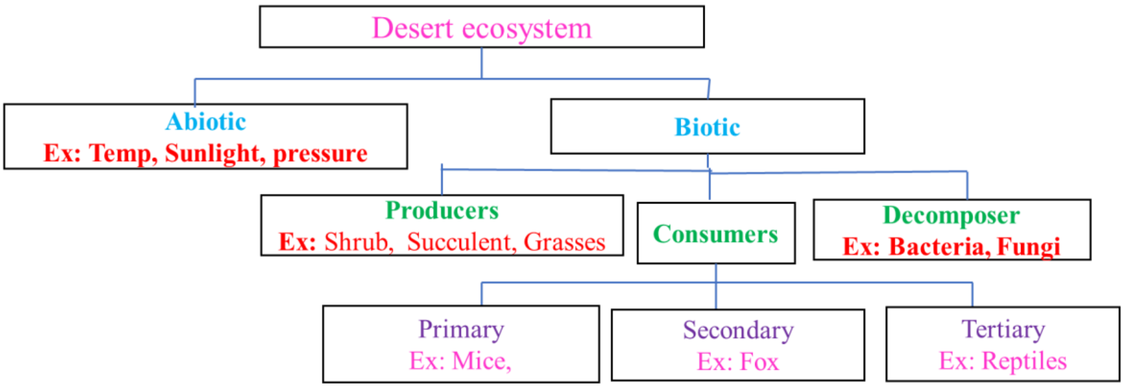
The biotic components include:

- Producers are shrubs, especially bushes, some grasses and a few trees.
- Consumers include animals like reptiles and insects which are able to live under xeric conditions. In addition, there are some nocturnal rodents and birds. The camel feeds on tender shoots of the plants.
- Decomposers are very few, as due to poor vegetation the amount of dead organic matter is correspondingly less. They are some fungi and bacteria.

Characteristics:

- The desert air in dry and the climate is hot.
- Characterized by less than 25% rain fall.
- Vegetation is very less. Because the soil is very poor in nutrients and organic matter.

Structure & functions of Desert ecosystem:



Aquatic ecosystem:

An aquatic ecosystem is an ecosystem that exists in water. Within an aquatic ecosystem, the environment is a watery one. Aquatic ecosystems can be divided into freshwater ecosystems (such as fresh rivers or freshwater lakes) and marine ecosystems such as the sea and rock pools.

1. Ponds ecosystem:

A Pond as a whole serves a good example of freshwater ecosystem.

Abiotic Components: The chief components are heat, light, P^H of water, CO_2, oxygen, calcium, nitrogen, phosphates, etc.

Biotic Components: The various organization that constitute the biotic component are as follows,

Producers: These are green plants, and some photosynthetic bacteria. The producer fix radiant energy and convert it into organic substances as carbohydrates, protein etc.

Macrophytes: these are large rooted plants, which include partly or completely submerged hydrophytes, eg Hydrilla, Trapha, Typha.

Phytoplankton: These are minute floating or submerged lower plants eg algae.

Consumers: They are heterotrophs which depend for their nutrition on the organic food manufactured by producers.

Primary Consumers: – Benthos: These are animals associated with living plants, detrivores and some other microorganisms.

Zooplanktons: These are chiefly rotifers, protozoans, they feed on phytoplankton.

Secondary Consumers: They are the Carnivores which feed on herbivores, these are chiefly insect and fish, most insects & water beetles, they feed on zooplanktons.

Tertiary Consumers: These are some large fish as game fish, turtles, which feed on small fish and thus become tertiary consumers.

Decomposers: They are also known as microconsumers. They decompose dead organic matter of both producers and animal to simple form. Thus, they play an important role in the return of minerals again to the pond ecosystem, they are chiefly bacteria, & fungi.

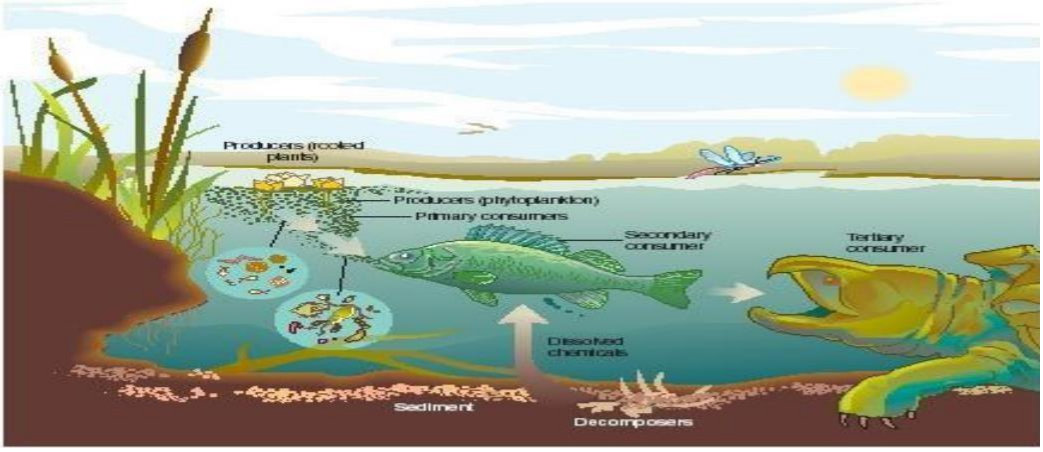
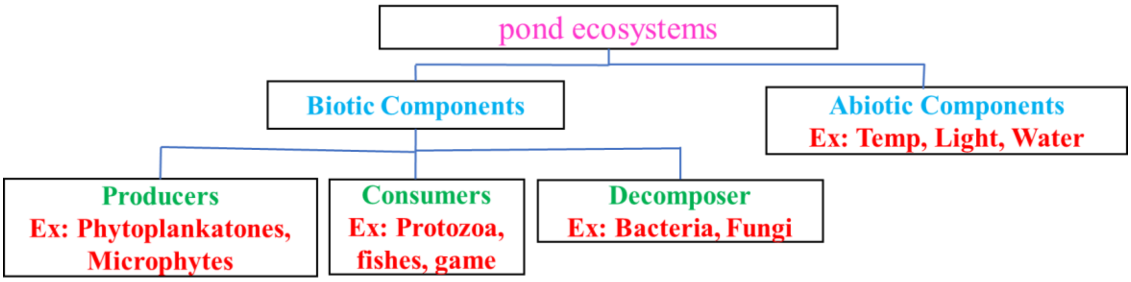


Fig.15: Ponds ecosystem

Characteristics:

- It is a small fresh water aquatic ecosystem.
- Pond is temporary, only seasonal.
- It is stored or stagnant water body
- Ponds get polluted easily due to limited amount of water.

component & function of pond ecosystems.



2. River or Stream ecosystem:

As Compared with lentic freshwater (Ponds & lakes), lotic waters such as streams, and river have been less studied. However, the various components of a riverine and stream ecosystem can be arranged as follows.

Producers: The chief producers that remain permanently attached to a firm substratum are green algae as Cladophora, and aquatic mosses.

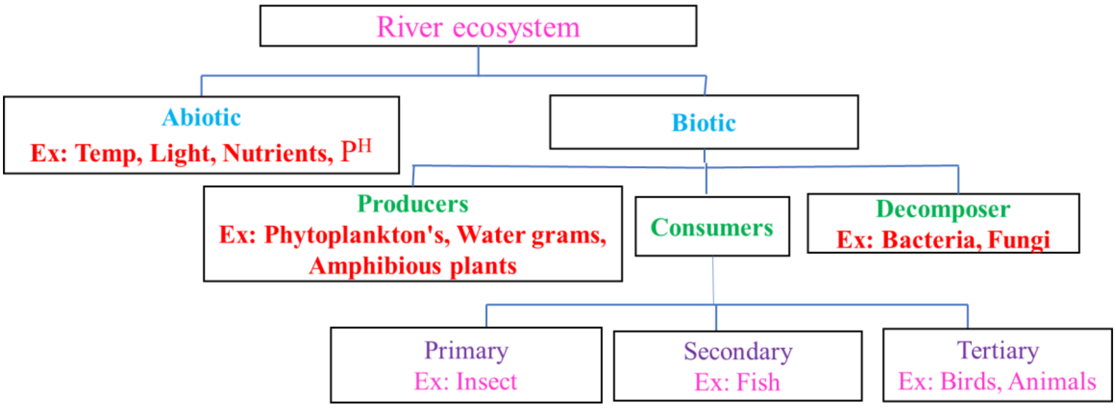
Consumers: The consumers show certain features as permanent attachment to firm substrata, presence of hooks & suckers, sticky undersurface, streamline bodies, flattened bodies. Thus, a variety of animal are found, which are fresh spongy and caddis-fly larvae, snails, flat worms etc.

Decomposers: Various bacteria and fungi like actinomycetes are present which acts as decompose.

Characteristics:

- ✓ It is a fresh water, and free flowing water system.
- ✓ Due to mixing of water, dissolved oxygen content is more.
- ✓ River deposits large amount of nutrients.

Structure & functions of River or Stream ecosystem:



3. Ocean ecosystem:

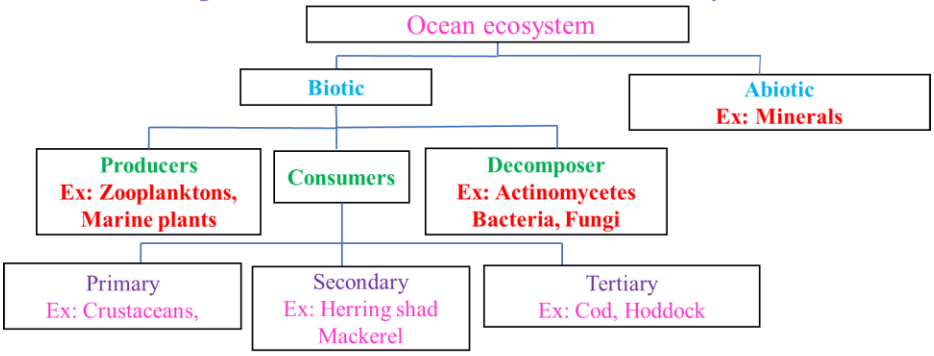
Ocean ecosystem is more stable than pond ecosystem, they occupy 70 % of the earth surface.

Abiotic Components: Dissolved oxygen, light, temperature, minerals.

Biotic Components:

Producers: These are autotrophs and are also known Primary producers. They are mainly, some microscopic algae (phyto-planlanktons) besides them there are mainly, seaweeds, as brown and red algae also contribute to primary production.

Components and function of an ocean ecosystem:



Consumers: They are all heterotrophic macro consumers.

Primary Consumer: The herbivores, that feed on producers are shrimps, Mollusks, fish, etc.

Secondary Consumers: These are carnivores' fish as Herring, Shad, Mackerel, feeding on herbivores.

Tertiary Consumers: These includes, other carnivores' fishes like, COD, Halibut, Sea Turtle, Sharks etc.

Decomposers: The microbes active in the decay of dead organic matter of producers, and animals are chiefly, bacteria and some fungi.

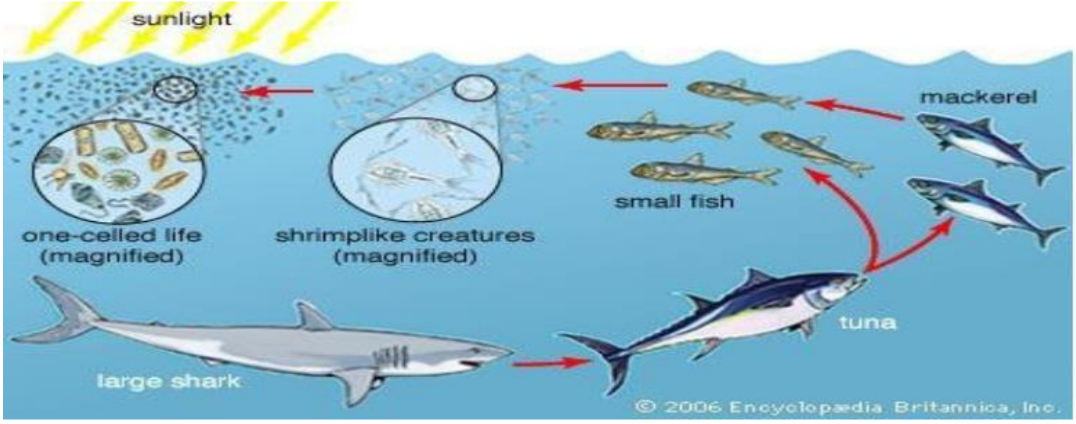


Fig.16: Ocean ecosystem

Characteristics:

- ✓ It occupies a large surface area with saline water.
- ✓ It is rich in biodiversity.
- ✓ It moderates the temperature of the earth.
- ✓ It is used for commercial like: Marin, ship etc.

4. Estuarine Ecosystem:

An estuary is a partially enclosed body of water along the coast where fresh water from river and streams meet and mix with salt water from oceans. These Ecosystems are considered as most fertile ecosystem.

Components and function of an Estuarine ecosystem:

- Estuarine ecosystem
  - Biotic
    - Producers
      Ex: March grasses, Sea weeds, Sea grams
    - Consumers
      Ex: Oysters, Crabs, Sea birds, Small fishes
    - Decomposer
      Ex: Actinomycetes Bacteria, Fungi
  - Abiotic
    Ex: Temp, Na, K salts, P^H

Abiotic Components: Nutrients such as phosphorus and nitrogen, temperature, light, salinity, P^H. This ecosystem experiences wide daily and seasonal fluctuations in temperature and Salinity level because of variation in freshwater in flow.

Biotic Components:

Producers: Phyplanktons- these micro-organisms manufacture food by photosynthesis and absorb nutrients such as phosphorous and nitrogen, besides them, mangroves, sea grass, weeds, and salt marshes.

Consumers: Primary consumers, Zooplanktons that feed on Phytoplankton, besides them some small microorganisms that feed on producers.

Secondary Consumer: Include worms, shellfish, small fish, feeding on Zooplanktons.

Tertiary Consumer: Fishes, turtles, crabs, starfishes feeding on secondary consumers.

Decomposers: Fungi & Bacteria are the chief microbes active in decay of dead organic matter.

Characteristics:

- ✓ Estuaries are transition zones, which are strongly affected by tides of the sea.
- ✓ Water characteristics periodically changed.
- ✓ The living organisms in Estuarine has wide tolerance.
- ✓ Salinity remains highest during the summers and lower in winters.

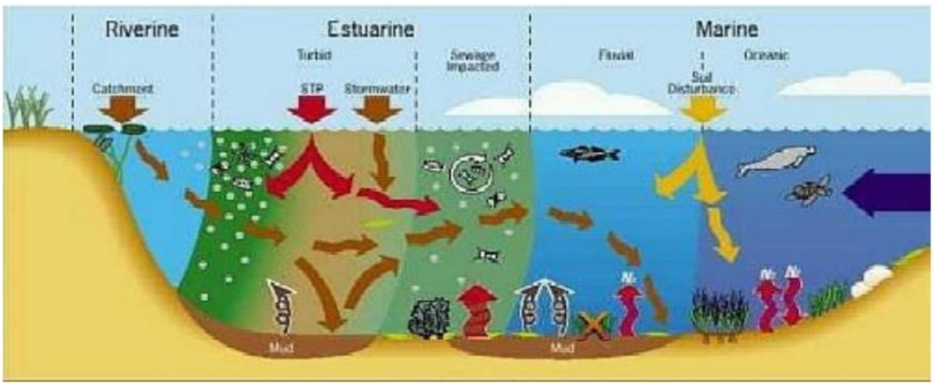


Fig.17: Estuarine ecosystem

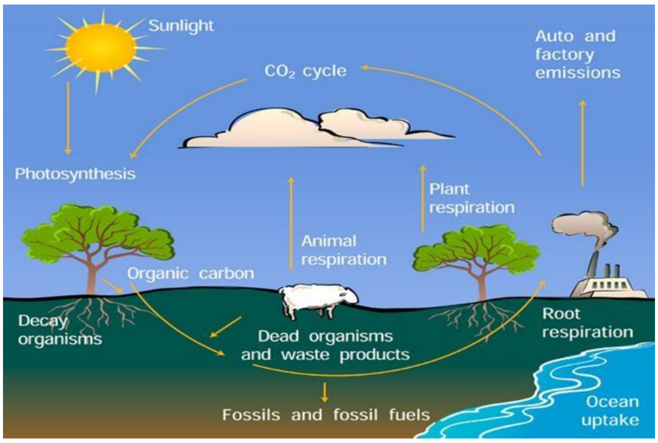
Bio Geo Chemical cycles

The cyclic flow of nutrients between the biotic and abiotic Component is known as nutrient cycle or bio- geo chemical cycle.

Nutrients: The elements, which are essential for the survival both plant and animals called nutrient. The Major are nutrients like C, H, O and N are cycled again and again between biotic and abiotic components in the ecosystem.

1. The Carbon cycle

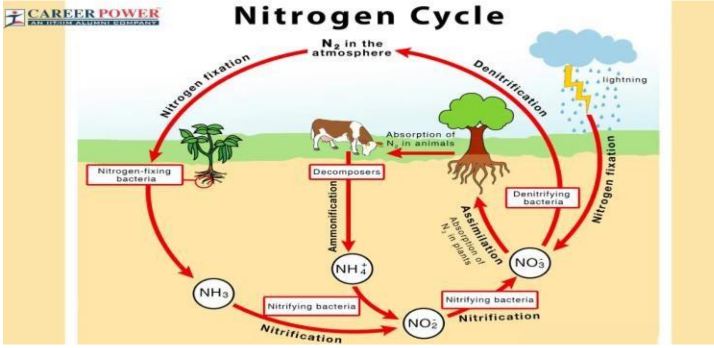
The physical cycle of carbon through the Earth's biosphere, geosphere, hydrosphere and atmosphere that includes such processes as photosynthesis, decomposition, respiration and carbonification. The carbon cycle describes the flow of carbon between the biosphere, the geosphere, and the atmosphere, and is essential to maintaining life on earth. Atmospheric Carbon Dioxide: Carbon in the earth's atmosphere exists in two main forms: carbon dioxide and methane. Carbon dioxide leaves the atmosphere through photosynthesis, thus entering the terrestrial and marine biospheres. Carbon dioxide also dissolves directly from the atmosphere into bodies of water (oceans, lakes, etc.), as well as dissolving in precipitation as raindrops fall through the atmosphere. When dissolved in water, carbon dioxide reacts with water molecules and forms carbonic acid, which contributes to ocean acidity. Human activity over the past two centuries has significantly increased the amount of carbon in the atmosphere, mainly in the form of carbon dioxide, both by modifying ecosystem's ability to extract carbon dioxide from the atmosphere and by emitting it directly, e.g. by burning fossil fuels and manufacturing
concrete.



Herbivorous animals feed on plant material, which is used by them for energy and for their growth. Both plants and animals release carbon dioxide during respiration. They also return fixed carbon to the soil in the waste they excrete. When plants and animals die, they return their carbon to the soil. These processes complete the carbon cycle.

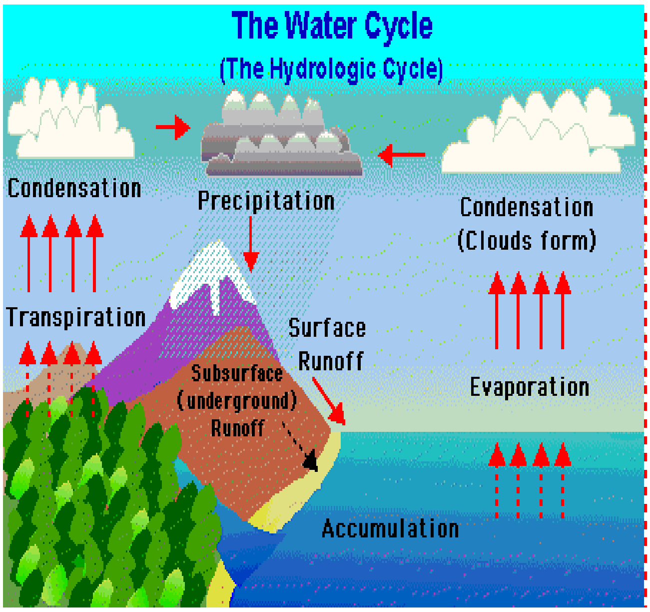
2. The Nitrogen Cycle:

Carnivorous animals feed on herbivorous animals that live on plants. When animals defecate, this waste material is broken down by worms and insects mostly beetles and ants. These small 'soil animals' break the waste material into smaller bits on which microscopic bacteria and fungi can act. This material is thus broken down further into nutrients that plants can absorb and use for their growth. Thus, nutrients are recycled back from animals to plants. Similarly, the bodies of dead animals are also broken down into nutrients that are used by the plants for their growth. Thus, the nitrogen cycle on which life is dependent is completed. Nitrogen fixing bacteria and fungi in soil gives this important element to plants, which absorb it as nitrates. The nitrates are a part of the plant's metabolism, which help in forming new plant proteins. This is used by animals that feed on the plants. The nitrogen is then transferred to carnivorous animals when they feed on the herbivores. Thus, our own lives are closely interlinked to soil animals, fungi and even bacteria in the soil. When we think of food webs, we usually think of the large mammals and other large forms of life. But we need to understand that it is the unseen small animals, plants and microscopic forms of life that are of great value for the functioning of the ecosystem.



3. The Water Cycle

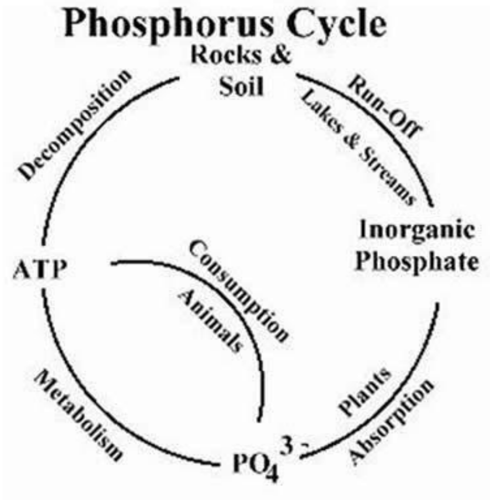
Water, air, and food are the most important natural resources to people. Humans can live only a few minutes without oxygen, less than a week without water, and about a month without food. Water also is essential for our oxygen and food supply. Plants breakdown water and use it to create oxygen during the process of photosynthesis.



The **water** (or hydrologic) **cycle** shows the movement of water through different reservoirs, which include oceans, atmosphere, glaciers, groundwater, lakes, rivers, and biosphere. Solar energy and gravity drive the motion of water in the water cycle. Simply put, the water cycle involves water moving from oceans, rivers, and lakes to the atmosphere by evaporation, forming clouds. From clouds, it falls as precipitation (rain and snow) on both water and land. The water on land can either return to the ocean by surface runoff, rivers, glaciers, and subsurface groundwater flow, or return to the atmosphere by **evaporation** or **transpiration** (loss of water by plants to the atmosphere).

An important part of the water cycle is how water varies in salinity, which is the abundance of dissolved ions in water. The saltwater in the oceans is highly saline, with about 35,000 mg of dissolved ions per liter of seawater. **Evaporation** (where water changes from liquid to gas at ambient temperatures) is a distillation process that produces nearly pure water with almost no dissolved ions. As water vaporizes, it leaves the dissolved ions in the original liquid phase. Eventually, **condensation** (where water changes from gas to liquid) forms clouds and sometimes precipitation (rain and snow). After rainwater falls onto land, it dissolves minerals in rock and soil, which increases its salinity. Most lakes, rivers, and near-surface groundwater have a relatively low salinity and are called freshwater. The next several sections discuss important parts of the water cycle relative to fresh water resources.

4. The Phosphorus Cycle:



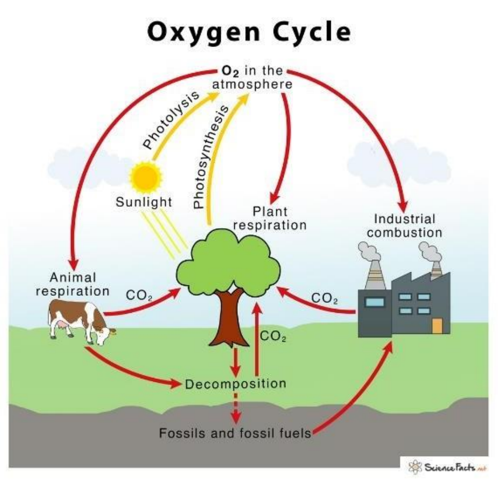
The constant exchange of a phosphorus or phosphates through living and nonliving things called **phosphorus cycle.**

Mineralization: Conversion of organic phosphorus into Inorganic phosphates called **Mineralization**. The enzymes involved in these reactions are collectively called as **phosphatases**.

Solubilization: Conversion of insoluble Inorganic phosphates into Soluble inorganic Phosphates. The soluble phosphate is taken by the plants. The availability of phosphorus depends on the degree of Solubilization various organic and inorganic acids produced by microorganisms like fungi, pencillium etc. in the soil.

5. The Oxygen Cycle:

The oxygen cycle is the cycle that helps move oxygen through the three main regions of the Earth, the Atmosphere, the Biosphere, and the Lithosphere. The Atmosphere is of course the region of gases that lies above the Earth's surface and it is one of the largest reservoirs of free oxygen on earth. The Biosphere is the sum of all the Earth's ecosystems. This also has some free oxygen produced from photosynthesis and other life processes. The largest reservoir of oxygen is the lithosphere.



Most of this oxygen is not on its own or free moving but part of chemical compounds such as silicates and oxides. The atmosphere is actually the smallest source of oxygen on Earth comprising only 0.35% of the Earth's total oxygen. The smallest comes from biospheres. The largest is as mentioned before in the Earth's crust. The Oxygen cycle is how oxygen is fixed for freed in each of these major regions.

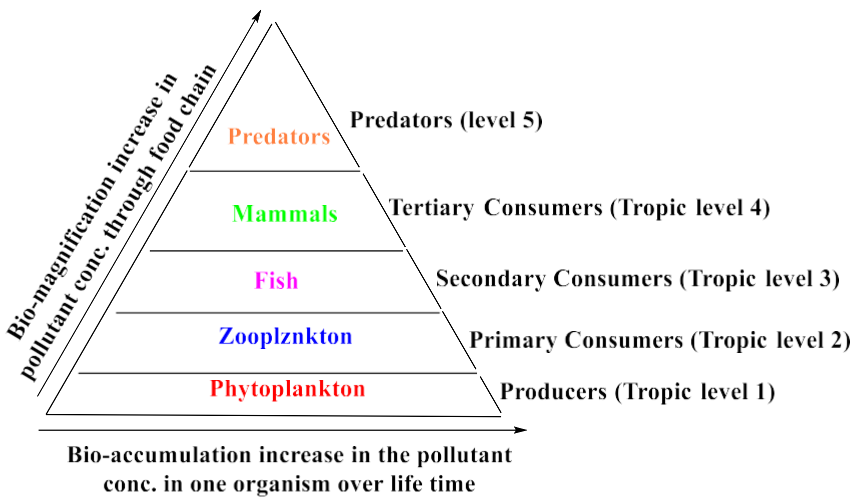
In the atmosphere Oxygen is freed by the process called photolysis. This is when high energy sunlight breaks apart oxygen bearing molecules to produce free oxygen. One of the most well-known photolysis it the ozone cycle. O_2 oxygen molecule is broken down to atomic oxygen by the ultra violet radiation of sunlight. This free oxygen then recombines with existing O_2 molecules to make O_3 or ozone. This cycle is important because it helps to shield the Earth from the majority of harmful ultra violet radiation turning it to harmless heat before it reaches the Earth's surface.

In the biosphere the main cycles are respiration and photosynthesis. Respiration is when animals and humans breathe consuming oxygen to be used in metabolic process and exhaling carbon dioxide. Photosynthesis is the reverse of this process and is mainly done by plants.

Bio magnification:

Bioaccumulation: The accumulation of a substance like pesticide (DOT), methyl mercury of other organic chemicals in an organism or part of an organism (tissue) is called bioaccumulation.

Bioaccumulation refers to the gradual build-up of pollutants in living organisms.



Biomagnification: Also Known as bio-amplification (or) biological magnification.

It is the increase in concentration of a pollutant that occurs in a food chain as a consequence of:

(1) persistence

(2) Bioenergetics in the food chain

(3) low rate of internal degradation / excretion of the substance often due to water-insolubility.

example: case of DDT

- When an animal consumes food having DDT residue, the DDT accumulates in the tissue of the animal by a process called bio accumulation.
- The higher an animal is on the food chain (e.g., tertiary consumers such as seals), the greater the concentration of DDT in their bodies as a result of a process called biomagnification.
- The higher an animal is on the food chain (e.g., tertiary consumers such as seals), the greater the concentration of DDT in their bodies as a result of a process called biomagnification.

Causes of Bioaccumulation / Biomagnification:

1. **Agricultural Products:** Chemicals from herbicides and pesticides penetrate the soil and accumulate to toxic levels. These chemicals can percolate into water bodies through surface runoff, eventually entering the food chain.
2. **Organic Contaminants:** Organic substances such as fertilizers released from agricultural farms contain harmful compounds. These substances are absorbed by plants and subsequently accumulate in organisms, transferring through various trophic levels.
3. **Plastic Pollution:** Disposal of plastic waste near or in water bodies leads to contamination. Additionally, ghost nets (fishing nets that have been abandoned in oceans) contribute to the problem.
4. **Mining:** Substances like zinc, copper, cobalt, and lead released from mining activities contaminate aquatic and nearby agricultural environments. These toxic metals are absorbed by aquatic organisms and crops, increasing toxicity levels significantly.
5. **Toxic Gases and Air Pollution:** Exhaust gases from vehicles and industries that manufacture or refine oil release pollutants into the air. These pollutants dissolve in rainwater (acid rain) and can contribute to biomagnification when they reach soil and water bodies.

Effects on Human Health

- The accumulation of mercury and PAHs (polycyclic aromatic hydrocarbons) in tissues and marine organisms poses serious health risks.
- The assimilation of heavy metals and toxic substances may lead to long-term effects such as kidney failure, respiratory disorders, brain damage, birth defects, and heart diseases.

Effects on Aquatic Animals

- The ingestion and subsequent accumulation of metals in the tissues of marine organisms have adverse effects on their development and reproduction.
- The consumption of heavy toxic metals by seabirds negatively affects egg production.

Disruption of the Food Chain

- The accumulation of substances that cause biomagnification can disrupt the natural food chain.

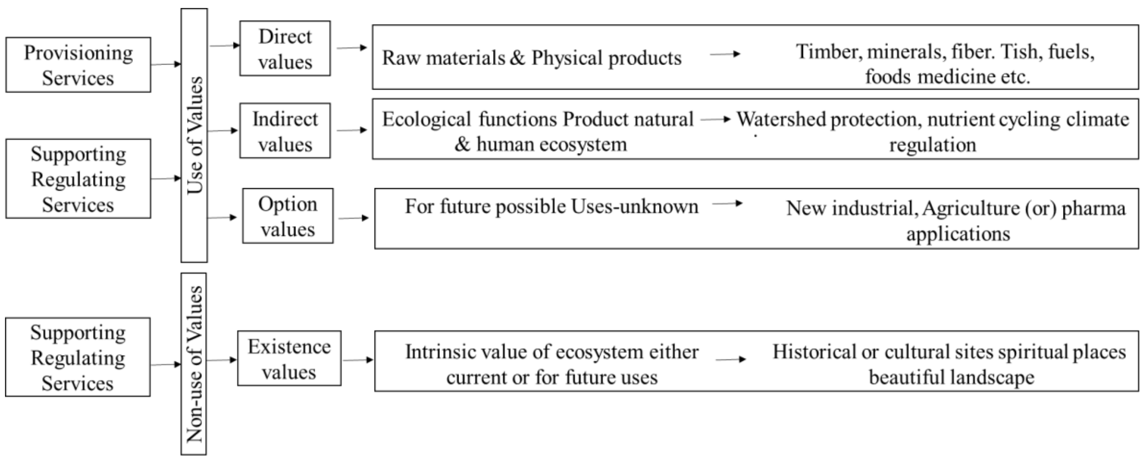
Destruction of Coral Reefs

- Cyanide used in gold leaching and illegal fishing practices leads to the destruction of coral reefs.

Control Measures:

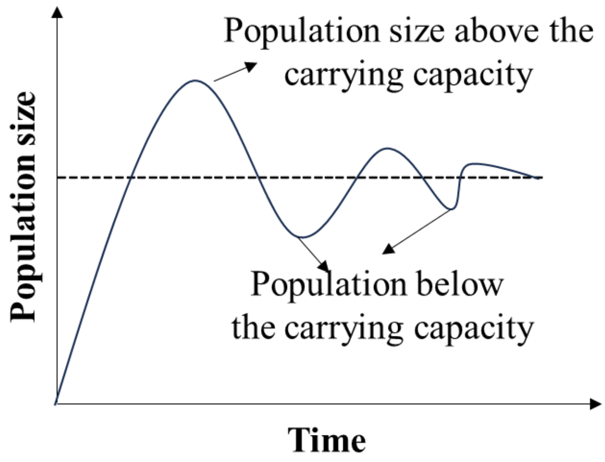
1. Eliminate heavy metals at the source.
2. Clean up areas contaminated with solid waste thoroughly.
3. Restrict the dumping of waste into rivers.
4. Prevent the washing of clothes and animals in rivers to reduce the inflow of chemicals from detergents.
5. Educating people and making them aware about the side effects.
6. Avoid usage of PVC.

Ecosystems value, Services and carrying capacity:



Carrying Capacity:

- **Carrying capacity** is the maximum number of individuals of a species that an ecosystem can support sustainably.
- It depends on the availability of resources such as food, water, and space.
- **Carrying capacity** refers to the maximum number of organisms that can live in a particular area without degrading the environment.
- A **carrying capacity graph** can be used to predict changes in population size:
- If the population rises **above** the carrying capacity, it will **decrease** due to resource limitations.



- If the population is **below** the carrying capacity, it may **increase** until balance is reached.
- Over time, the population tends to **stabilize at the carrying capacity**.
- **Limiting Factors:** A limiting factor is any condition that causes a population to stop growing or limits its growth.

  Abiotic Factors: These are the physical components of an ecosystem, such as temperature, P^H, rainfall, etc.

  Biotic Factors: These are the biological influences on organisms, such as vegetation, competition, and predation.